Lesson 8 Proof by contradiction

Prop 1 There is no largest integer.

Proof Suppose there is a largest integer, say n, then $n+1$ is an integer larger than n, contradicts that n is a largest integer.

negation

condiction!

Symbol: $\rightarrow\leftarrow$

Prop 2 Consecutive integers cannot be both even.

Direct proof: Let x and $x+1$ be (any) consecutive integers.

If x is not even, then we are done.

If x is even, then $x = 2n$, $n \in \mathbb{Z}$, and $x+1 = 2n+1$, which odd and not even.

Proof by contradiction Suppose there are consecutive integers x and $x+1$, which are both even. Then $x = 2n$ and $x+1 = 2m$, and $x+1 = 2n+1 = 2m$.

So, $2n+1 = 2m$, which

implies that $1 = 2m - 2n = 2(m-n)$
and 1 is even, which is absurd!
Latin: reductio ad absurdum (false)
reduction to absurdity

Prop 3 $\sqrt{2}$ is irrational.
Proof Suppose $\sqrt{2}$ is rational,
then $\sqrt{2} = \frac{m}{n}$, where m, n are
integers which do not share a
common factor other than ± 1, i.e.
$\frac{m}{n}$ is an irreducible fraction.

$$\sqrt{2} = \frac{m}{n} \rightarrow 2 = \frac{m^2}{n^2}$$

$$\rightarrow 2n^2 = m^2$$

$$\rightarrow m^2 \text{ is even}$$

$$\rightarrow m \text{ is even}$$

(Prop 1, lesson 7)

$$\rightarrow m = 2k,\ k \in \mathbb{Z}$$

$$\rightarrow m^2 = (2k)^2 = 4k^2$$

$$2n^2 = 4k^2 \rightarrow n^2 = 2k^2$$

$\to n^2$ is even

$\to n$ is even

So, n and m share a common factor, 2, contradicting that $\frac{m}{n}$ is irreducible.

Prop 4 There are infinitely many primes.

Proof Suppose there are only finitely many primes, $p_1, p_2, \ldots, p_t$.

(We assume the Fund Thm of Arithmetic: every integer greater than 1 has a prime factor.)

$0 \le 1 < p_i$

Let $x = p_1 p_2 \cdots p_t + 1$. Then x

remainder

does not have a factor among $p_1, p_2, \ldots, p_t$ because there is always a remainder of 1 when x is divided by p_1 or $p_2, \ldots$ or p_t. Since $x > 1$ and does not have prime factor, this contradicts FTA.

Prop 5 $a, b \in \mathbb{R}$. If $\underset{}{\overset{p}{ab=1}}$, then $\overset{q}{a \leq 1 \text{ or } b \leq 1}$.

Last time: contrapositive $\sim q \to \sim p$

If $a > 1$ and $b > 1$, then $ab > a(1) > 1$

$ab \neq 1$.

$\sim(p \to q) = p \wedge \sim q$

Contradiction: Suppose $ab = 1$, and $a > 1$ and $b > 1$. Then $ab > a(1) > 1$ and $ab \neq 1$, contradicting $ab = 1$.

$p \to q$: do contrapositive over contradiction